Subject
of
Mini Project
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Topic
of
Online Coding Platform

BACHELOR OF ENGINEERING
in
INFORMATION TECHNOLOGY

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1. Aim of the Project

- The aim of the project is to develop an Online Coding Platform with AI-assisted guidance that helps technical students enhance their problem-solving and interview preparation skills. The system provides a unified platform for DSA practice, AI-based coding assistance, 1-vs-1 coding challenges, and interview simulation, enabling users to practice coding efficiently, track progress, and prepare for technical roles through an interactive and user-friendly environment.

2. Project Scope

- The Online Coding Platform is a web-based system designed to support students preparing for technical interviews and competitive programming. The platform integrates multiple learning and assessment features such as problem-solving, AI guidance, and live challenges.

- This project is designed for individuals seeking:

    1. Practice of Data Structures and Algorithms (DSA) problems.
    2. AI-assisted guidance while solving coding problems.
    3. 1-vs-1 coding challenges with friends.
    4. Interview preparation and live interview simulation for technical roles.
    5. Progress tracking and performance analysis.

3. Project Objective

- The primary objective of the Online Coding Platform is to develop an intuitive, web-based system that enables users to:

    1. Solve coding problems: Users can practice coding problems across different difficulty levels and programming languages.
    2. AI-assisted coding support: The platform provides context-aware AI guidance to help users understand problem statements, optimize solutions, and learn better approaches.

3. Interview preparation: The system offers AI-based interview practice and live interview simulation to prepare users for real technical interviews.

4. Project Modules

- The online coding platform consists of multiple interconnected modules, each responsible for delivering specific functionality:

1. Authentication Module
2. Problem Management Module
3. Interview Practice Module
4. One-on-one Coding Challenge Module
5. Progress Tracking Module
6. Profile Section Module

5. Project Basic Requirements:

1.5.1 Hardware:

- **Processor**: Minimum Intel Core i5 or equivalent
- **RAM**: 4 GB or higher
- **Internet Connection**: Stable internet connection

1.5.2 Software:

- **Frontend**: HTML & CSS, React.
- **Backend**: Node.js, Express.js.
- **AI Integration**: AI models for code assistance and interview simulation. (Ollama : gpt-oss model)
- **Database**: MongoDB.
- **Hosting**: AWS.

6. Comparison of Existing Applications with your Project / Literature review:

6.1 Comparison:

Table 6.1 Comparison Table

Feature	Online Coding Platform	Leetcode (Existing system)
Primary Functionality	• Coding practice and interview preparation with AI Assistance.	• Coding practice focused mainly on DSA and competitive programming
Interactivity	• Interactive problem solving with AI guidance and interview preparation simulation	• Interactive problem solving without interview simulation
Interview Preparations	• Interview preparation based on user's job description and role requirements	• No job-specific interview preparation
Coding Challanges	• One-vs-One coding challenges to compete with friends	• Weekly and contest-based challenges (not direct 1-vs-1)
Progress Tracking	• Detailed performance analytics with personalized feedback	• Limited AI features

Table 6.2 Comparison Table

Feature	Online Coding Platform	Hacker Rank (Existing system)
Primary Functionality	• Coding practice and interview preparation with AI Assistance.	• Coding practice and skill assessment mainly for hiring tests
Interactivity	• Interactive problem solving and interview preparation simulation. • Available interview preparation based on user's requirement of job description.	• Interactive problem solving without personalized interview simulation
Interview Preparations	• AI and Live Interview for job specific roles.	• No live or AI-driven interview simulations
Coding Challanges	• One vs One coding challenge available for competing between friends.	• Competitive challenges, but no direct 1-vs-1 friend mode.
Progress Tracking	• Detailed performance analytics	• Detailed Performance analytics

6.2 Literature Review:

i. Integration of Generative AI in Programming Education.

Abstract

- Integrating generative artificial intelligence tools like ChatGPT and GitHub Copilot into programming education presents notable opportunities to enhance learning but also raises critical challenges in balancing innovation with preserving foundational programming skills. The Findings reveal that successful integration hinges on intentional teaching strategies, thoughtfully designed assessments, and structured integration processes. However, barriers such as GenAI tools' limited accessibility features, insufficient bias mitigation, and a tendency to prioritize tool availability over curriculum alignment often disrupt seamless adoption. Additionally, the potential for students to become over-reliant on AI risks diminishing higher-order thinking skills and programming logic. The findings emphasized the necessity of strategic implementation, educator training, and inclusive practices to maximize GenAI's potential in programming education.

Summary

- While GenAI tools like ChatGPT and Codex streamline code generation and foster engagement, they can hinder the development of higher-order thinking skills and understanding of programming logic if applied without structured learning activities. Other limitations that hinder the seamless integration include accessibility barriers, cultural biases, and educator readiness, emphasizing the necessity of inclusive policies and professional development. However, while GenAI-Ped offers a structured approach from an objective setting to evaluation, it still requires empirical validation through the classroom to confirm its broad applicability.

References : https://link.springer.com/article/10.1007/s40593-025-00524-3

ii. Artificial intelligence in education (AIED): a roadmap to a future research agenda

Abstract

- Despite the increased adoption of Artificial Intelligence in Education (AIED), several concerns are still associated with it. This has motivated researchers to conduct (systematic) reviews aiming at synthesizing the AIED findings in the literature. However, these AIED reviews are diversified in terms of focus, stakeholders, educational level and region, and so on. This has made the understanding of the overall landscape of AIED challenging. To address this research gap, this study proceeds one step forward by systematically meta-synthesizing the AIED literature reviews. Additionally, when discussing AIED, strong focus was on higher education, where less attention is paid to special education. The results also reveal that AI is used mostly to support teachers and students in education with less focus on other educational stakeholders (e.g. school leaders or administrators). The study provides a possible roadmap for future research agenda on AIED, facilitating the implementation of effective and safe AIED.

Summary

- The field of AI is advancing rapidly and is integrated into different domains of research and practice every day. Due to the potential benefits and the ability to transform teaching and learning, AI has become increasingly discussed, adopted and researched. Such rapid increase has been obvious in the multitude of studies being carried out worldwide. Therefore, several literature reviews have been conducted to map the existing literature from different perspectives. The fragmented and varied landscape and scope of existing literature reviews motivated us to perform an overarching systematic review that synthesizes all existing systematic reviews to offer a grand perspective of the state of the art. We further enhance our review with a future research agenda based on our analysis and synthesis of existing studies. This comprehensive coverage allows us to have a broad perspective as well as to devise a research agenda in form of recommendations, challenges and research directions.

References : https://link.springer.com/article/10.1186/s40561-024-00350-5

iii. Design of an Online Programming Platform and a Study on Learners' Testing Ability.

Abstract

- Online judge (OJ) systems are essential in programming education as they efficiently evaluate learners' programming skills and reduce instructor workload. However, these systems often overlook the importance of software-testing concepts. To address this gap, we developed a system called Pytutor that integrates software-testing concepts to assess learners' programming abilities and proficiency by exploring students' test cases and learning behaviors. Drawing on software engineering theory and practical techniques, test capabilities are evaluated by analyzing the code coverage and mutation testing of Defining Test Cases. Since our experiment is conducted in an online environment, we can collect students' learning behaviors and further analyze the relationship between software engineering abilities and learning behaviors. We also analyzed the differences in programming and testing abilities between computer science majors and non-computer-science majors. Our findings suggest that better testing abilities may contribute to the improvement in programming abilities, whereas in the current Taiwanese education context, computer science majors do not necessarily have better testing abilities. This result provides suggestions for us to strengthen software-testing education no matter which type of students it is targeted at.

Summary

- In our research, we observed that learners with higher testing abilities can better solve programming problems during the learning process. Based on the findings, we recommend that programming-education practitioners consider incorporating software-testing concepts into course design—whether or not learners are computer science majors. This approach allows learners to receive education in both programming fundamentals and software testing, ultimately leading to improved learning outcomes. Although this result may not have a huge impact on the education sector, it can be used as a reference for cultivating students' software engineering abilities.

References : https://www.mdpi.com/2079-9292/12/22/4596

iv. An Exploratory Analysis of Feedback Types Used in Online Coding Exercises.

Abstract

- Online coding environments can help support computing students gain programming practice at their own pace. Especially informative feedback can be beneficial during such self-guided, independent study phases. This research aims at the identification of feedback types applied by CodingBat, Scratch and Blockly. Tutoring feedback as coined by Susanne Narciss along with the specification of subtypes by Keuning, Jeuring and Heeren constitute the theoretical basis. Accordingly, the five categories of elaborated feedback (knowledge about task requirements, knowledge about concepts, knowledge about mistakes, knowledge about how to proceed, and knowledge about meta-cognition) and their subtypes were utilized for the analysis of available feedback options. The study revealed difficulties in identifying clear-cut boundaries between feedback types, as the offered feedback usually integrates more than one type or subtype. Moreover, currently defined feedback types do not rigorously distinguish individualized and generic feedback. The lack of granularity is also evident in the absence of subtypes relating to the knowledge type of the task. The analysis thus has implications for the future design and investigation of applied tutoring feedback. It encourages future research on feedback types and their implementation in the context of programming exercises to define feedback types that match the demands of novice programmers.

Summary

- This study analyzes feedback provided by online coding platforms such as CodingBat, Scratch, and Blockly to support self-paced programming practice. It finds that feedback often combines multiple types, lacks clear distinction between generic and personalized feedback, and needs better structure to meet the learning needs of novice programmers.

References : https://arxiv.org/abs/2206.0

v. A web-based programming learning environment to support cognitive development.

Abstract

- Web-based programming has become a popular and vital issue in recent years. The rapid growth of various applications not only demonstrates the importance of web-based programming, but also reveals the difficulty of training relevant skills. The difficulty is owing to the lack of facilities such as online coding, debugging and peer help to assist the students in promoting their cognitive development in web-based programming. To cope with these problems, in this paper, a web-based programming assisted system, "WPAS", is proposed, which is able to support five programming activities with various difficulty levels of cognition based on Bloom's cognitive taxonomy. WPAS provides online coding, debugging and annotation tools to conduct the training and peer assessment for web-based programming. Experimental results of 47 undergraduate students show that the innovative approach is helpful to the students in improving their cognitive development in Web-based programming. In addition, according to the results of the questionnaire, most of the participants perceived the ease of use and usefulness of the proposed system. Therefore, this study suggests that teachers could design Web-based programming activities supported by the WPAS system to improve students' cognitive development in web-based programming.

Summary

- This study proposes a web-based programming assisted system (WPAS) that supports online coding, debugging, annotation, and peer assessment based on Bloom's cognitive taxonomy. Experimental results with undergraduate students show that the system effectively improves cognitive development in web-based programming and is perceived as easy to use and useful.

References : https://www.sciencedirect.com/science/article/abs/pii/S095354380800045

7. Project Feasibility Study:

- **Technical Feasibility:**

  - The proposed Online Coding Platform can be developed using modern and well-established frontend technologies such as HTML, CSS, and React, which support the creation of responsive and interactive user interfaces.
  - Backend functionality can be efficiently implemented using Node.js and Express.js, which are widely used for building scalable and high-performance web applications.
  - MongoDB is suitable for storing user data, coding problems, submissions, interview records, and progress tracking information due to its flexibility and scalability.
  - AI-assisted guidance and interview simulation can be integrated using AI models (Ollama: gpt-oss model), enabling features such as code explanation, solution optimization, and interview question generation.
  - The platform can support online coding practice, problem evaluation, and real-time interactions using RESTful APIs and secure server-side execution mechanisms.
  - Therefore, the project is technically feasible using currently available technologies and development resources.

- **Operational Feasibility:**

  - The Online Coding Platform provides a unified environment for coding practice, AI-assisted learning, interview preparation, and competitive challenges.
  - A user-friendly interface ensures that students can easily navigate the problem, solve problems, and access features without any prior training.
  - Features such as authentication, problem categorization, one-on-one coding challenges, interview simulations, and progress tracking improve overall usability and engagement.Chatbots are intuitive, hence user is not required to provide any training.
  - The system supports self-paced learning, allowing users to practice coding and prepare for interviews according to their individual schedules.
  - Since similar platforms are already widely accepted by students, the proposed system is expected to be readily adopted by the target audience. Hence, the project is

operationally feasible and meets the needs of technical students and job aspirants.

- **Economic Feasibility:**

  - The development cost of the project is relatively low as it uses open-source frameworks and libraries such as React, Node.js, and Express.js.
  - AI integration using Ollama (gpt-oss model) reduces dependency on costly third-party AI services.
  - Cloud hosting services like AWS (free or low-cost tiers) can be used for deployment, minimizing infrastructure expenses during the initial phase.
  - No specialized hardware is required beyond standard development systems with basic internet connectivity.
  - Therefore, the project is economically feasible with minimal development and operational costs.

Overall Feasibility

- The Online Coding Platform is technically, operationally, and economically feasible.
- The project leverages readily available technologies, requires moderate development effort, and incurs low initial costs.
- With proper implementation, the system can effectively support coding practice, AI-guided learning, interview preparation, and skill development for technical students.
- Overall, the project is practical, scalable, and suitable for real-world deployment.

8. Project Timeline Chart:

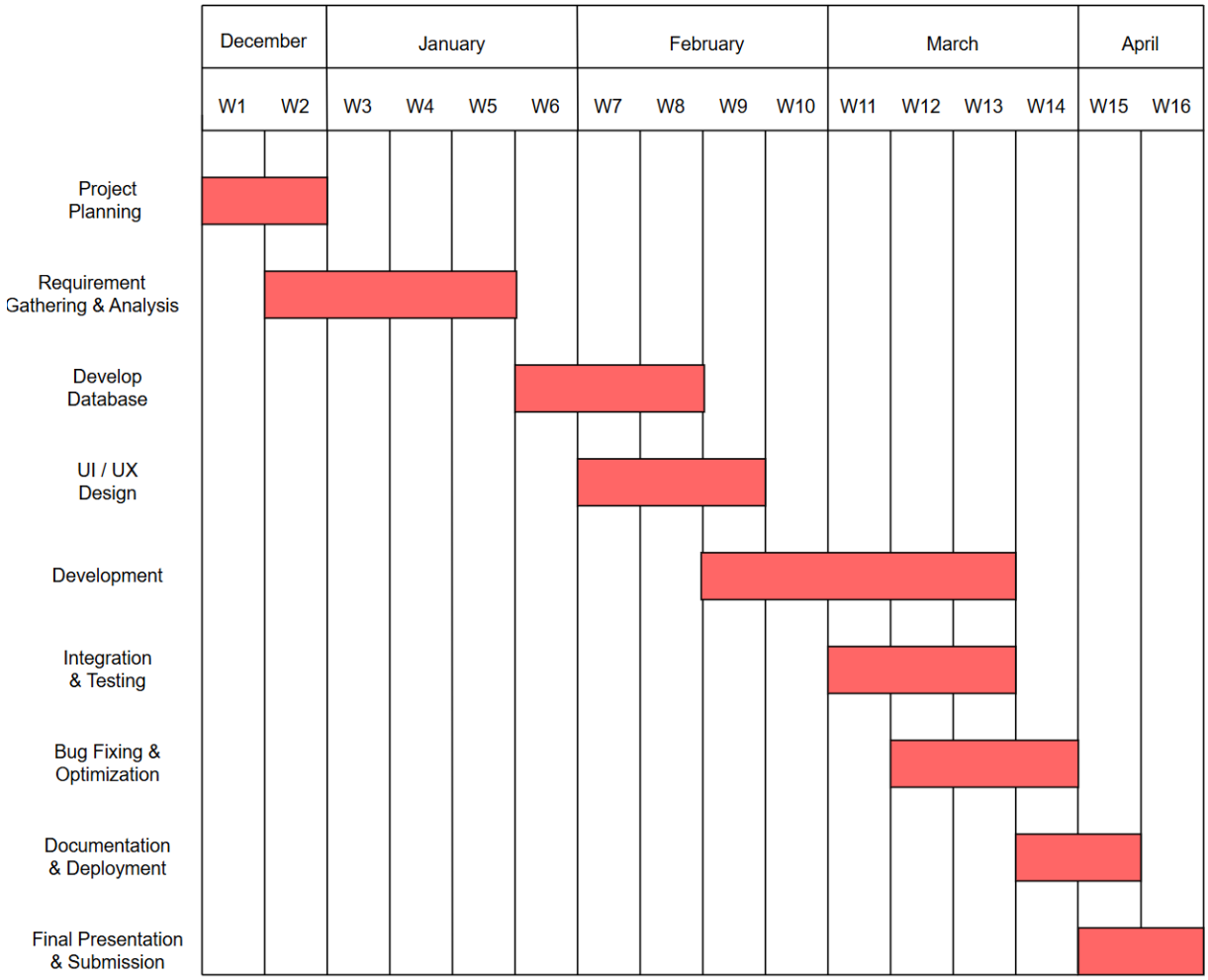

Figure 2.1 Project Timeline Chart

9. Detailed Modules Description:

- The Online Coding Platform with AI-assisted Guidance consists of multiple interconnected modules, each designed to provide specific functionality and collectively deliver an interactive, efficient, and user-friendly coding and interview preparation experience. The primary modules include:

1. Authentication Module:

- Manages secure access to the platform and user identity verification.
- **User Registration:** Allows new users to create accounts using email and password credentials.
- **User Login & Logout:** Authenticates registered users and manages secure session handling.
- **Session Management:** Maintains active user sessions using tokens to ensure secure access.
- **Authorization Control:** Restricts access to platform features based on authentication status.

2. Problem Management Module:

- Handles the storage, organization, and evaluation of coding problems.
- **Problem Repository:** Maintains a structured database of DSA problems with descriptions, constraints, and examples.
- **Difficulty Classification:** Categorizes problems into Easy, Medium, and Hard levels.
- **Test Case Management:** Stores visible and hidden test cases used to evaluate user submissions.
- **Code Submission Handling:** Manages code submissions and forwards them for compilation and execution.
- AI Assistance Service:
  - Provides intelligent, context-aware support to improve learning and problem-solving.
- **AI-Based Code Guidance:** Assists users with hints, explanations and optimization

suggestions while solving problems.

- **Error Detection & Explanation:** Analyzes syntax and logical errors and explains them in an understandable manner.
- **Interview Question Assistance:** Generates interview-style questions and follow-up discussions using AI models.

3. Interview Practice Module:

- Supports technical interview preparation through AI-driven interview simulations.
- **Live AI Interview**: Conducts voice-based technical interviews to simulate real interview interactions.
- **DSA Interview Round:** Presents Data Structures and Algorithms problems similar to real technical interview rounds.
- **Timed Sessions:** Applies time constraints to replicate real interview pressure.

4. One-on-One Coding Challenge Module:

- Enables competitive and collaborative coding experiences.
- **Live Coding Challenges:** Allows two users to participate in real-time coding challenges.
- **Problem Synchronization:** Ensures both participants receive the same problem and constraints.
- **Real-Time Progress Tracking:** Displays code updates, execution results and completion status.
- **Result Comparison:** Compares performance based on correctness, execution time and efficiency.

5. Progress Tracking Module:

- Monitors and analyzes user activity and performance.
- **Submission History:** Maintains a record of all attempted and solved problems.
- **Performance Analytics:** Displays statistics such as accuracy, problem-solving trends, and difficulty-wise progress.
- **Interview Performance Tracking:** Tracks results from interview simulations and challenges.
- **Visualization:** Presents progress through graphs and summaries for better understanding.

6. Profile Section Module:

- Manages user-specific data and achievements.
- **User Profile Management:** Allows users to view and update personal information.
- **Skill Overview:** Displays problem-solving skills and topic-wise strengths.
- **Achievements & Badges:** Rewards users for milestones such as problem completion and consistent activity.
- **Public Profile:** Enables users to showcase their coding progress and performance.

10. Project Work Distribution

- Jagdish Pathakji
  - Problem Management Module
  - One vs One Coding Challenge Module
- Yug Dholakiya
  - Authentication Module
  - Interview Practice Module
- Rudra Gohil
  - Progress Tracking Module
  - Profile Section Module

Remarks	Signature